Experiments

Biology

S.No.	Title
1.	Comparative study of atmospheric carbon dioxide and that which is released during respiration.
2.	Testing the effect of saliva on starch.
3.	Identifying the gas released during photosynthesis.
4.	Identifying male and female reproductive organs in the given flower.

Chemistry

S.No.	Title
1.	Study of neutralization reaction.
2.	Study the effect of air and moisture on iron.
3.	Identifying washing soda and baking soda from given chemical samples.
4.	Study esterification reaction between alcohols and alkanoic acids.

Physics

S.No.	Title
1.	Experimentally compare Celsius and Fahrenheit scale and verify the relation between them.
2.	Verify Ohm's law.
3.	Establishing laws of reflection using a plane mirror.
4.	Establishing laws of refraction or finding the refractive index of rectangular glass slab.
5.	Finding the focal length of a concave mirror using one-pin method.
6.	Drawing magnetic force lines by aligning the North Pole of a bar magnet with geographic north.
7.	Drawing magnetic force lines by aligning the North Pole of a bar magnet with geographic south and finding neutral point.

Practical Work

Biology

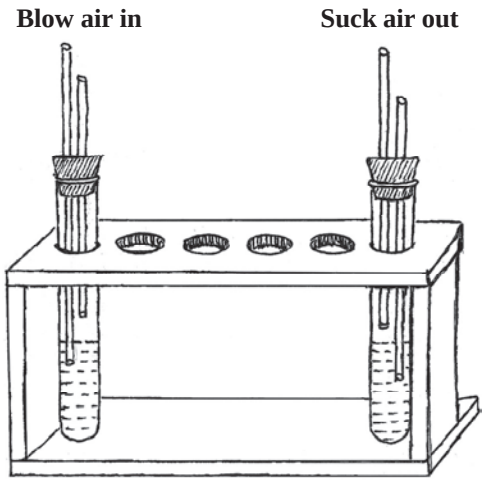
Practical work Life Process: Nutrition, Respiration, Transport, Excretion Respiration

Aim: To make a comparative study of carbon dioxide in exhaled air during breathing and atmospheric air.

Materials Required: Two boiling tubes, a test tube stand, lime water(freshly prepared),two straws/ empty refill (nib removed)/two glass tubes, rubber corks with two holes, a little wheat dough/plasticine, watch

Method:

1. Arrange the apparatus as given in the figure. Use plasticine or wheat dough to fix the straw/refill of glass tube into the cork so that the apparatus is air tight.
2. Label the two boiling tubes as A and B respectively.
3. Fill lime water in the experimental set up to the height shown in the figure.
4. Blow into boiling -tube A. Suck air from B.
5. Note the time taken for lime water to turn milky in A and in B.



Observation: You would find that lime water turns milky faster in A than in B. We know that presence of CO_2 tested in this way. Your observation indicates that there was more CO_2 in exhaled air than in inhaled air(or atmospheric air) as more CO_2 would turn lime water milky sooner.

Result: It may thus be concluded that CO_2 in exhaled air is more than that in atmospheric air.

Precautions:

1. If you keep on blowing air in lime water the milkiness would disappear. So stop blowing as soon as the lime water turns milky.
2. The straws/glass tubes should be dipped in lime water as shown in figure.
3. Wash apparatus before and after use.

Instruction: Use freshly prepared lime water by dissolving a tube of lime(can be acquired from betel leaf/paan shops) in about 150 ml of distilled water. Keep the solution for about an hour; decant the clear liquid and use as lime water.

Practical work Life Process: Nutrition, Respiration, Transport, Excretion Respiration

Aim: To observe the effect of saliva on starch

Materials required: Two beakers, two test tubes, a test tube stand, a teaspoon full of wheat flour, distilled water, dropperand iodine solution(preferably tincture iodine).

Procedure: Add half a teaspoon of wheat flour to half a beaker(around 125 ml) of distilled water mix well to prepare a solution. Now take a little of this solution in a test tube add a few drops of tincture iodine and see if it turns blue black or not that is gives confirmatory test for presence of starch . In case it doesn't, add more of wheat flour till you get a confirmatory test for starch. Now start your experiment with this solution.

Take two test tubes and label them as A and B respectively. Keep them in a test tube stand. Add 20-25 drops of wheat flour solution to both test tubes.

Now to test tube A add a similar amount of your saliva. Do not add anything to test tube B. Leave the set ups for over an hour.

Add iodine solution to both test tubes A and B. Observe which one shows presence of starch.

The teacher may ask following questions to infer.

Questions:

1. What happened after adding iodine to both test-tubes A and B?
2. Why do you think this may have happened?
3. What do you think is the effect of saliva on wheat flour solution?

Result: Saliva helps in breakdown of starch to other compounds that do not test positive for starch test with iodine. Salivary amylase in saliva is mainly responsible for the process of breakdown.

Precautions:

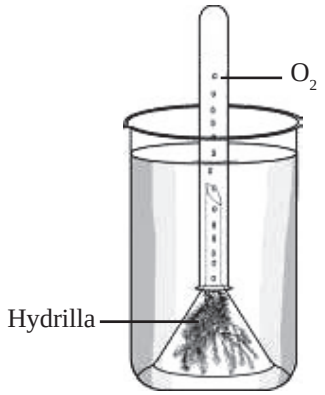
1. Add same amount of wheat flour solution to both test tubes.
2. Wash your mouth thoroughly before adding saliva to test tube A the amount.

Practical work Life Process: Nutrition, Respiration, Transport, Excretion Respiration

Aim: To observe the effect of sunlight on plants and to test the gas released.

Materials Required: Two beakers, two funnels, two test-tubes, (Cheela) /elodea, black paper or cloth,incense sticks and match box.

Procedure: Arrange two sets of the experimental set up as shown in the figure. The set ups have to be arranged under water (use the bucket full of water for the same). Ensure that the test-tube as well as funnel are full of water at the start of your experiment. Cover one set up with black paper and keep it in shade (ensure that it gets minimum heat and light) Let this be labeled as B. Keep the other set up,



labeled as A in sunlight. Leave the set ups as such for a minimum of 3-4 hours. Keep observing from time to time. You would find bubbles rising up and gas collecting in the test-tube. Once the test tube of the set up kept in sunlight is around half filled with gas, take the whole set up under water and remove the test tube and close its mouth with your thumb. Ensure that the gas does not escape. Now bring the test tube out of water, keep a burning incense stick ready and as soon as you remove your thumb, insert the incense stick into the test-tube. Did the incense stick start burning more brightly? Which gas do you think was collected in the test-tube? In which of the set ups, A or B was the collection of this gas more?

Observation:

1. More gas was collected in A. This showed that light was necessary for collection of the gas.
2. The gas present was oxygen. Presence of which is tested by using a burning incense stick /a burning splinter.

Result: We find that Oxygen is evolved in the presence of sunlight.

Precautions:

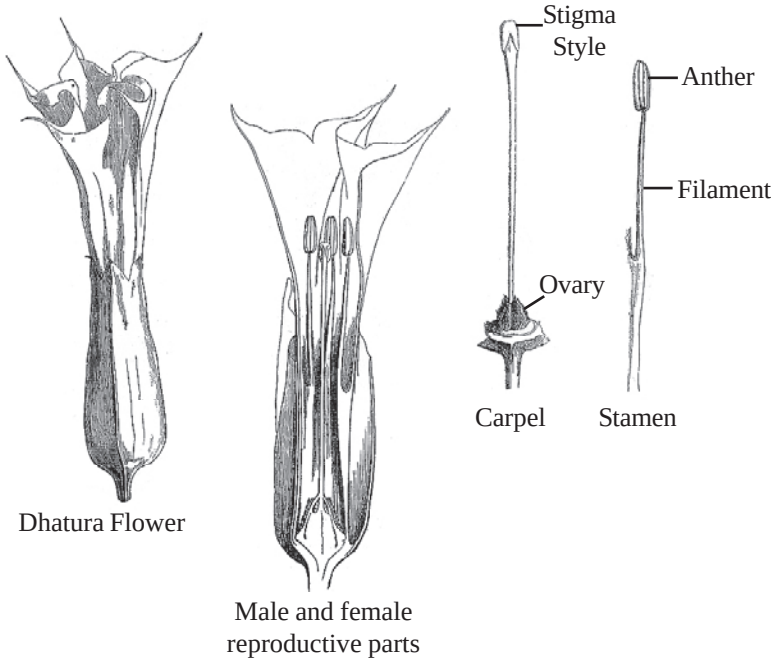
1. The set up kept in shade should be covered with black paper/ cloth such that entry of light may be minimized.

   Set up has to be arranged under water carefully such that there are no water bubbles in the set up initially..
2. Burning incense stick has to be inserted into test-tube quickly.

Practical work Life Process: Growth, Development and Reproduction

Aim: To observe reproductive parts in a flower

Materials Required: A flower of dhatura or railway creeper, slide, brush, forceps, needle, petridish/ watch glass/bowl, coverslip, hand lens, glycerine.



Dhatura Flower

Male and female reproductive parts

Procedure: Take the flower in your hand and open it up (refer to the section of practical work for figure). Now observe the female and male reproductive parts. Figure in this section shows a Dhatura flower and the male and female reproductive organs in it.

You may observe these organs as under-

You would have to remove the outer whorls as shown in the figure.

(i) Androecium: This is the male reproductive organ of a flower, composed of all the stamens present in the flower. Each stamen has a head formed of anther bearing pollen in pollen sacs and a thin connector called filament that attaches the anther to a petal, thalamus(top part of stalk of flower) etc. The anther of Dhatura appears as arrow heads with fine powdery structures on it called as pollen.Pollens are structures that pass on to the stigma of the female reproductive part. The process of transfer of pollen from anther to stigma is called as pollination. If recognized on stigma they germinate to reach the ovule where the fusion of male and female nuclei takes place. Plants may be self- pollinated when pollen of a flower reaches the stigma of the same flower or other flower on the same plant. Cross pollination occurs when pollen from flower on one plant reaches stigma of flower on another plant and is recognized there.

(ii) Gynoecium: The female reproductive organ of a flower is composed of all the carpels. A carpel is mainly composed of three parts called as stigma, style and ovary (see the figure).

Ovary: This is the lowest part of a carpel where ovules or female reproductive cells or gametes are present. Ovules after fertilization develop into seeds that can give rise to future generation of the plant.

Take longitudinal and transverse sections of the ovary, stain with safranin, wash excess stain with water, prepare slides of the same and observe under the microscope. Note your observation in your copy.

Precaution:

1. Use blade or cutter carefully during section cutting.
2. Wash excess stain before preparing a slide. Do not allow air bubbles to accumulate under the coverslip.

Chemistry

Experiment (Acid, Base and Salt)

Objective : Study of neutralization reaction

Essential apparatus: Test tubes, test tube stand, dropper, dilute hydrochloric acid, dilute sodium hydroxide and phenolphthalein

Theory : Acids and alkalis react with one another to form salt and water and this reaction is known as neutralization.

Acid + Alkali ⟶ Salt + Water

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

When all acid molecules have reacted with the alkali then the solution becomes neutral. An indicator is used to detect neutralization point. In this experiment, if phenolphthalein is used as indicator then it will be colourless in acidic medium and pink in basic medium.

Procedure : Take 20 drops of dilute hydrochloric in a test tube and add 1-2 drops of phenolphthalein to it. With the help of a dropper, add dilute sodium chloride drop by drop to the above solution while continuously shaking the test tube. Keep adding sodium hydroxide till the colour of solution in the test tube become light pink. Calculate the number of drops of dilute sodium hydroxide solution added and note the reading in the observation table. Repeat this process two more times.

Observation Table

S.No.	Drops of dilute hydrochloric acid taken	Drops of dilute sodium hydroxide solution needed for neutralization
1.	20 drops	.....
2.	20 drops	.....
3.	20 drops	.....

Conclusions : Neutralization of 20 drops of dilute hydrochloric acid required ……drops of dilute sodium hydroxide solution.

Precautions : 1. Test tube should be clean and dry.

2. All solutions should be prepared in distilled water.

3. Clean the dropper with water each time before use.

Experiment (Metals and metallurgy)

Objective : Study of the effect of air and moisture on iron

Essential apparatus: 3 Test tubes, test tube stand, awl-pins, cotton, 3 balloons

Theory : Presence of factors such as air and moisture is necessary for rusting of iron.In this process, the surface of iron acts like an electrochemical cell. One part of the surface acts as anode and another part acts as the cathode. Moisture (or humidity) - which contains dissolved oxygen, carbon dioxide - acts as an electrolyte. The electrochemical process of rusting of iron can be shown as follows:

At the anode: Fe goes into the solution as Fe2+

$$Fe(s) \longrightarrow Fe^{2+} + 2e^-$$

At the cathode: In the presence of oxygen, these electrons are taken up water molecules to form OH- ions.

$$2H_2O + O_2 + 4e^- \longrightarrow 4OH^-$$

The Fe2+ ions formed on the anode react with the OH- to form Fe(OH)2(s). This iron hydroxide reacts with the oxygen in the atmosphere in the presence of moisture to give hydrated iron oxide.

$$2Fe(OH)_2 + \tfrac{1}{2} O_2(aq) + H_2O \longrightarrow Fe_2O_3.xH_2O(s)$$

This hydrated iron oxide is known as rust.

Procedure : Take three test tubes and label them 'A', 'B' and 'C' respectively. Put 3-4 awl-pins in each test tube. Add enough water in test tube 'A' so that the pins are submerged. Moisten a piece of cotton and put it in lowest part of test tube 'B'. Leave awl-pins in test tube 'C' untouched. Now make the test tubes air-tight by placing balloons on their mouths. Put the test tubes in a test tube stand and place them in a hot place (or sunlight) and observe after 3-4 hours.

Observation table

S.No.	Test tube	Observation prior to putting in sunlight	Observation after 3-4 hours in the sunlight	Conclusion
1.	Test tube 'A'	Awl-pin submerged in water		
2.	Test tube 'B'	Awl-pin and moist cotton		
3.	Test tube 'C'	Awl-pin and air		

Conclusions : The necessary conditions for rusting of iron are ……..and …….

Precautions :

1. This experiment should be carried out in the summers or winter. Presence of excess moisture during the rainy season will give to error in results.
2. Awl-pins should be completely clean. If rust is present on the, clean thoroughly with sandpaper before use.
3. Ensure that the test tube is completely dry and has no trace of moisture.
4. Balloons should be tightly and properly placed on the mouths of the test tubes.

Experiment (Chemistry in Daily Life)

Objective : Identifying baking soda or washing soda from given chemical samples.

Essential apparatus: sodium carbonate (washing soda) or sodium hydrogencarbonate (baking soda), sodium chloride, red litmus paper, lime water (freshly made), dilute hydrochloric acid, 4 test tubes, one boiling tube, test tube stand,

Theory : Washing soda ($Na_2CO_3.10H_2O$) and baking soda ($NaHCO_3$) are both white solids. Their aqueous solutions are alkaline and turn red litmus blue. Carbonates and hydrogencarbonates react with dilute acids to release carbon dioxide gas which turns lime water milky.

$$Na_2CO_3(s) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$$

$$NaHCO_3(s) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$$

$$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$$

Lime water *milky*

When excess of CO_2 is passed through lime water then calcium hydrogencarbonate is formed. It dissolves in water to give a colorless solution.

$$CaCO_3(s) + H_2O(l) + CO_2(g) \longrightarrow Ca(HCO_3)_2(aq)$$

Procedure :

1. In three test tubes, separately take a pinch of substance from each of the three given samples. Label the test tubes 'A', 'B' and 'C'.
2. Add 5 mL distilled water to each of the test tubes. Gently shake the test tubes.
3. Using a glass rod put a drop of solution 'A' on red litmus paper. Clean the glass rod with distilled water and do the same for samples 'B' and 'C'. Note the change of colour of litmus paper in all three cases.
4. Add 1 mL dilute hydrochloric acid to each of the test tubes. Did you observe slight bubbling in any of them? If yes, then test the sample which bubbled as described below.
5. To test using lime water, take the bubbling solution in a boiling tube and set up the apparatus as shown in figure-1.
6. Using a thistle funnel add dilute hydrochloric acid drop by drop to the boiling tube.

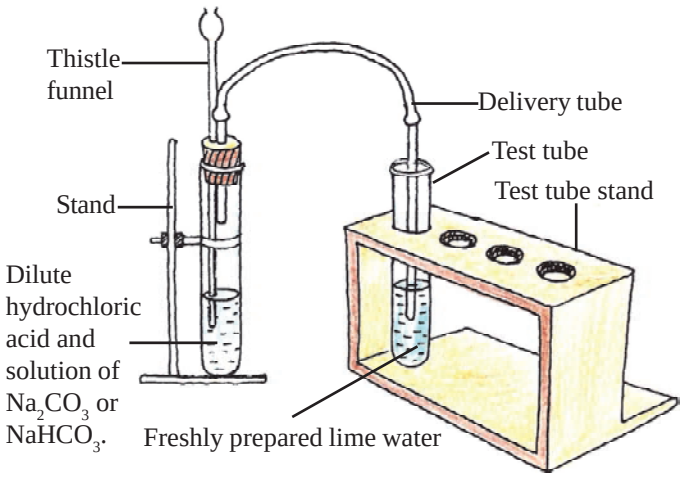


Figure-1: Identification of carbon dioxide gas using lime water

7. Pass the gas formed through freshly prepared lime water. Did the lime water turn milky? If yes, then it shows the presence of carbon dioxide.

8. Continue to pass the gas through lime water. Did it become colourless? If yes, then it confirms that the gas released is carbon dioxide.

Observation table

S.No.	Solution of sampl	Red litmus paper changes to blue or does not show any change	Gas released with slight bubbling on addition of dilute hydrochloric acid	Lime water turns milky or not
1.	'A'			
2.	'B'			
3.	'C'			

Conclusions : Washing soda/ Baking soda were identified from among the given samples based on observations with litmus paper and lime water.

The sample in test tube ……is baking soda/ washing soda.

Precautions :
1. Dilute hydrochloric acid should be added drop by drop otherwise a rapid reaction may take place and the reaction mixture could mix with the lime water.
2. While using washing soda and dilute hydrochloric acid, ensure that the samples do not touch your skin as they may harm you.
3. Use freshly prepared lime water to test for carbon dioxide.

Experiment (Derivatives of Hydrocarbons)

Objective : Study of esterification reaction between an alcohol and alkanoic acid.

Essential apparatus: 5 mL ethanoic acid, 5 mL ethanol, a few drops of concentrated sulphric acid, distilled water, 1 g sodium hydrogencarbonate, thermometer (-10°C to 100°C), test tubes, measuring cylinder (10 mL), 250 mL beaker, burner, tripod stand, wire gauze

Theory : When an alkanoic acid is heated with an alcohol in the presence of concentrated sulphuric acid, water and an ester is formed as product. The reaction is known as esterification, for example when ethanoic acid reacts with ethanol then ethyl ethanoate (ester) and water are produced. Esters have a distinct fruity smell which is different from the smell of acids and alcohols.

$$CH_3COOH + CH_3CH_2OH \xrightarrow[\text{On warming}]{\text{Conc. } H_2SO_4} CH_3COOCH_2CH_3 + H_2O$$

Ethanoic acid *ethanol* *ethyl ethanoate*

Procedure **:**

1. Take 5 mL acetic acid in a clean test tube and add 4-5 drops concentrated sulphuric acid to it.
2. Add 5 mL ethanol to the test tube.
3. Take 150 mL water in a beaker and warm it to 60oC. Remove the heating apparatus (spirit lamp) once the water is warm.
4. Place the test tube in the warm water for about 5 minutes so that the reaction mixture also becomes warm. This is known as heating in water bath (figure-2).
5. Keep shaking the test tube from time to time.
6. Pour the reaction mixture in the beaker containing the solution of sodium hydrogencarbonate in water. Doing so will cause the unreacted ethanoic acid to react with sodium hydrogen carbonate. Do you see any bubbles?
7. Try to feel the difference in smells of ethanoic acid, ester and ethanol.

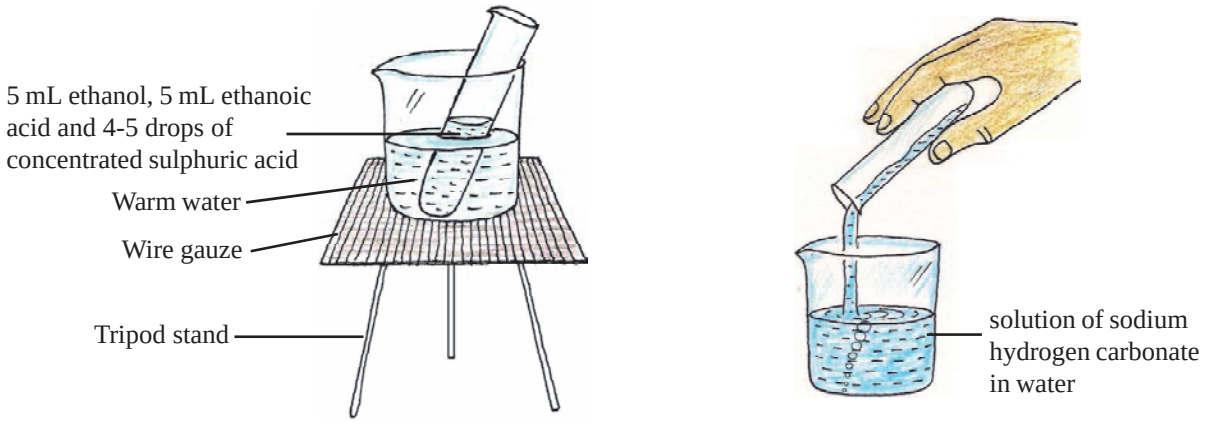


Figure-2: Reaction between an alcohol and alkanoic acid

Conclusions **:** Write about the difference in smell of ethanoic acid, ester and ethanol.

An ester is formed when the -OH group of an alcohol is replaced by -OR group (where -R denotes an alkyl group). Concentrated H_2SO_4 is used as a dehydrating agent in this reaction.

Precautions **:**

1. Be careful while using concentrated H_2SO_4.
2. Carbonic compounds vaporize easily and a water bath should be used for heating inflammable compounds such as alcohols. Never heat alcohol directly over a spirit lamp.

Physics
Some instruments used in experiments

1. Clinical Thermometer - The thermometer used by doctors to measure body temperature of patients is known as clinical thermometer.



Structure - It is a small thermometer. As shown in the figure there is a bulb which contains mercury. The capillary tube has a kink near the bulb which prevents the mercury from flowing back into the bulb. It has a glass stem on which readings are marked from 95°F to 110°F. Since human body temperature fluctuates between 95°F to 110°F, therefore this range is used in clinical thermometers. Normal body temperature, for a healthy person, is shown by a red arrow at 98.4°F.

Use - To use the thermometer, it is first washed with water and then given a quick jerk to bring down the level of mercury in the glass tem. After this, on coming in contact with the patient's body the mercury starts rising. Once the mercury becomes stable, the reading is noted. To use again, the entire process is repeated. Due to kink in the stem, the mercury does not come back on its own. To read the thermometer it should be held horizontally, parallel to our eyes so that the reading is visible.

Why mercury is used in thermometers?

1. Mercury expands uniformly with increase in temperature.
2. It does not stick to the walls of the glass tube.
3. Its specific heat is negligible therefore it needs minimal heat to show increase in temperature.
4. It has a freezing point of -38.83°C and boiling point of 356.7°C which gives a very big range to measure temperatures.
5. It shows uniform volume expansion therefore, it is convenient to mark readings.
6. It is very bright and reflective, so it is easy to read accurately.
7. It has very low vapour pressure so the impurities present are negligible and mercury is mostly pure.



These days, doctors use digital thermometers based on electronic principles. You will read more about them later.

2. Using a multimeter



You may have seen this instrument with nearly all technicians. Ammeter, voltmeter and ohmmeter are all combined in one instrument. This means that we can measure direct current (DC), alternating current (AC) and resistance using this instrument. It has a red needle and a black needle, called probes, which are used to connect the multimeter to the electric circuit.

The method of connecting the multimeter to the circuit depends on the range of values which have to be measured. If we want to use it as an ammeter

to measure current then it should be connected in series in the circuit. If potential has to be measured then it should be connected in parallel fashion in the circuit.

Voltmeters and ammeters have two ends; a red end which is the positive pole and a black end which is the negative pole. Similarly, multimeter also has two probes and the negative end is known as the common point. The black probe is always fitted in the common point. Where the red probe is fitted depends on what is to be measured.

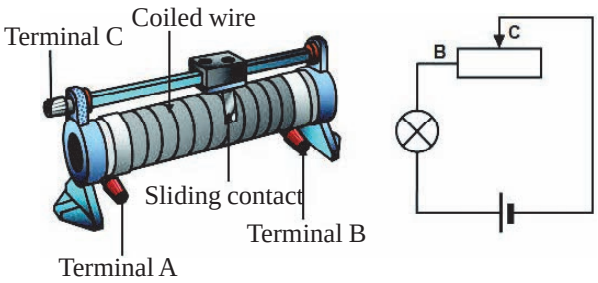
(A) Measuring electric currents - Multimeter has more than one range to measure any values. For example, the multimeter shown in the figure can measure DC in the following range: 2000 mA, 20 mA, 200 mA, 10 A. Choose any one range to fix the scale of multimeter which means that after we set a range, the multimeter will be able to measure values only between the maximum and minimum of the range.

If we have no idea about how much current is flowing through the circuit then take the biggest range in the beginning. If the current is low and we are not able to measure it accurately then move to a lower range.

(B) Measuring voltage/potential - It can measure both AC and DC voltage. The multimeter shown in the figure can measure DC voltage in the following range: 1000V, 200V, 20V, 2000 mV, 200 mV and also to measure AC voltage at 750 V and 200 V.

(C) Measuring resistance - The multimeter shown in the figure can measure in the range: 2000 kohms, 200 kohms, 20 kohms, 2000 ohms and 200 ohms.

3. Rheostat - It is made of an insulating material like ceramic shaped into a hollow cylinder core around which a high specific resistance wire like Manganin or Constantine is coiled as shown in the figure. The ends of wire are known as terminal A and terminal B.



It contains a metal rod on which a bar of conducting material is attached. Rheostat is used to change the value of electric current flowing through the circuit. The values of total resistance and maximum current that can flow through it are engraved on the coil.

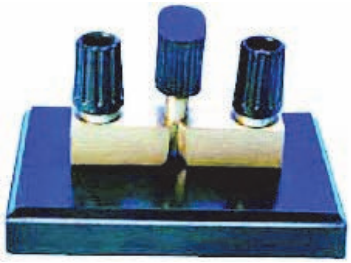
4. Resistance box - In this brass pieces or knobs are placed in two rows on an ebonite box. There is some empty space between any two knobs. The first and last knobs have connecting screws on them. Standard resistors are present beneath each knob.



Standard resistors are made of materials like Manganin, Eureko etc. Each end of the wire is connected to one resistor. The space between the brass knobs is plugged

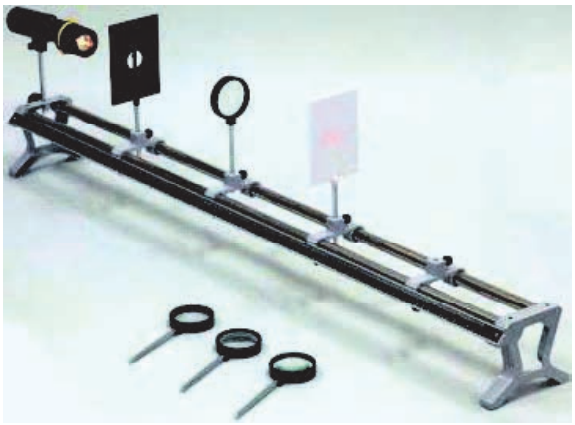
with brass pieces. For safety, the metal coils are placed in a wooden box. The resistance under each knob is indicated outside on the box. The box has resistances of 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000 ohms or 0.1, 0.2, 0.5, 1, 2, 5, 10 ohms. In most boxes infinity symbol (∞) is indicated on the knobs to which the wires are joined. In reality, these knobs are not connected by any wire.

5. One-way plug key - Is shown in the figure. In this key, two brass plugs appropriately spaced are attached to an ebonite surface. A brass plug which has a connecting screw is placed in the space.



When we wish that electric current should flow through the circuit, the third plug is inserted and it is removed to stop the flow. This key is universally used in electricity related experiments and functions like an on/off switch.

6. Optical bench - it is used in light related experiments. It is 1 m or longer bench made of steel or wood. On one side of the bench, a cm scale is marked and an inch scale is marked on the other side.



Different stands are attached to the bench on which mirrors and lenses can be fixed using a pin. This pin can be moved up and down as needed. The bench can be made to lie flat or inclined using a screw below it.

Experiment (Heat and temperature)

Objective : Experimentally compare Celsius and Fahrenheit scale and verify the relation between them.

Essential Apparatus: Beaker, Celsius and Fahrenheit thermometers, tripod, stand, stirrer, wire etc.

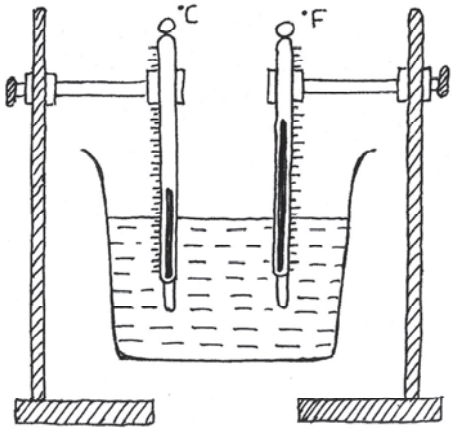
Theory : If we wish to convert the temperature from Celsius scale to and Fahrenheit scale then the relation (conversion formula) is as follows:

$$\frac{C}{5} = \frac{F-32}{9}$$

$$C = \frac{5}{9} \times (F-32)$$

Procedure :

1. Find the least count of the given Celsius and Fahrenheit thermometers.
2. Note the room temperature on Celsius and Fahrenheit thermometers. Use t1 to indicate the Celsius scale and t2 for Fahrenheit scale. While noting the temperature, be careful that your eyes are parallel to the highest point of the thermometer.



3. Now fill a small beaker about two-thirds with water and heat it (using a spirit lamp). Insert the two thermometers in the beaker as shown in the figure so that their bulbs are submerged completely. Make sure that your thermometers do not touch the bottom of the beaker. When the water starts boiling then agitate it using a stirrer so that the temperature becomes uniform throughout the water. Take the readings on the two thermometers; they show the boiling point of water.
4. Remove the lamp and allow the water to cool.
5. After the water has cooled a little, again agitate it with the stirrer and take **readings on the thermometer. Do this 3-4 times.**

Observation table : Least count of Celsius (centigrade) scale = 1°C

Least count of Fahrenheit scale = 1°F

S.No.		Celsius scale reading, t_1°C	Fahrenheit scale reading, t_2°C	$t_3 = \frac{9}{5}(t_1 + 32)$
1.	Room temperature	22	72	71.6
2.	Water temperature	99	210	210.2
3.				
4.				
5.				
6.				

Calculations : For each reading t_1 on the Celsius scale, we will calculate the corresponding t_3 using the formula given below and write it in the observation table.

$$t_3 = \frac{9}{5} \quad t_1 + 32$$

Results : (i) It is clear from the observation table that the boiling point of water = 99°C=210°F

(ii) For each observation, the value of t_3 is nearly same, and thus the following formula is verified.

$$\frac{C}{5} = \frac{F - 32}{9}$$

Precautions :

1. While taking the temperature of steam, the thermometer bulbs should not be in water but should be in the steam.
2. While taking the temperature of boiling water, the thermometer bulbs should be submerged in water.
3. Use the stirrer to agitate the water before taking the readings.
4. Both thermometers should be vertical.
5. Thermometer should not touch the beaker walls.
6. While taking the reading, our eyes should be parallel to the reading point.

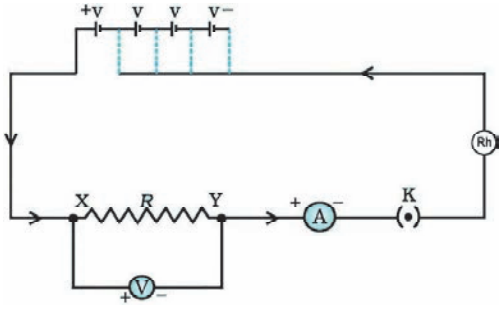
Experiment (Electric currents and circuits)

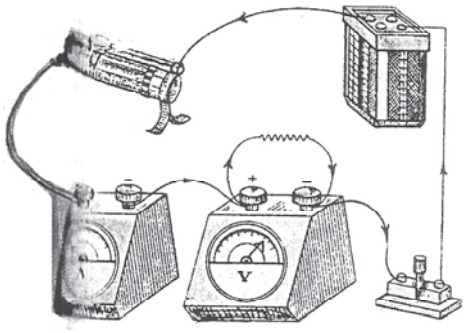
Objective : Verification of Ohm's Law.

Essential Apparatus: Resistor wires, cells, key, ammeter, voltmeter, connecting wires, variable resistor.

Theory : According to Ohm's law, provided that there is no change in physical state of the conductor, there is a linear relationship between the voltage drop (V) across its two ends and the current (I) flowing through it. This means that V = IR where R is the resistance of the conductor.

Procedure : 1. Find complete the circuit as shown in the figure. To do this we will connect the battery, resistor wires, ammeter, key and variable resistor in series and connect the voltmeter in parallel to the resistor wire. Make sure that the positive end of the battery is connected to positive ends of the ammeter and voltmeter.





2. Note the least count of the voltmeter and ammeter.
3. Now place the plug in the key and with the help of rheostat control the current flowing through the ammeter. We know that ammeter measure current and voltmeter the potential drop so let us note the respective readings.
4. use the rheostat to change the current and take several more readings for current and corresponding voltage.
5. Take 4-5 readings for current and voltage.

Observations : Least count of ammeter = 0.1 V

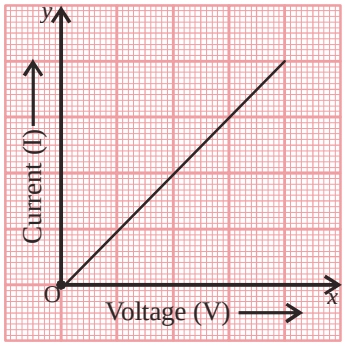
Least count of voltmeter = 0.1 A

S.No.	Ammeter reading (I)	Voltmeter reading (V)	$\frac{V}{I}$ resistance in ohms
1.			
2.			
3.			
4.			
5.			

Average resistance R = ...

Results : The ratio of voltage and current is constant. Therefore, we may conclude that the potential drop across the resistance wire is proportional to the current flowing through it.

Graph : Plot between V and I



Precautions : 1. Ammeter should always be joined in series in the circuit.

2. Voltmeter should always be joined in parallel in the circuit.
3. Do not constantly pass current through the conductor.
4. All screws should be tightened.
5. Do not pass very high currents through the conductor.

Experiment (Light: Reflection and refraction from a plane surface)

Objective : Verification of laws of reflection using a plane mirror.

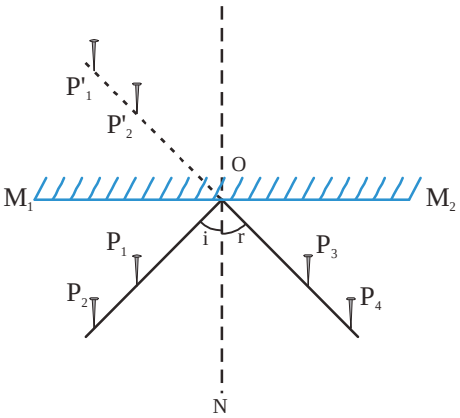
Essential Apparatus: A plane mirror fixed vertically in a plastic/ wooden frame, drawing board, drawing paper (White), a foot-rule, a pencil, pins, push pins, a protactor, etc

Principle : When a ray of light is incident on a reflecting surface then it travels back in the original medium after falling on the surface. This phenomenon is called reflection of light and it follows certain laws.

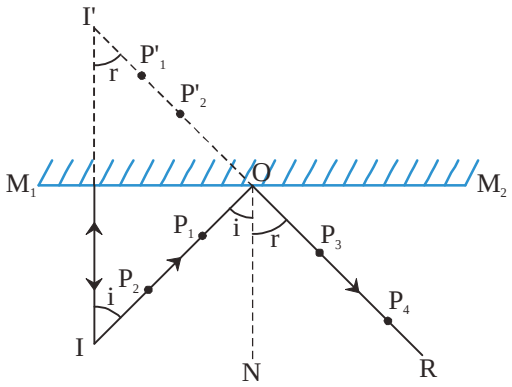
1. The incident ray, the reflected ray, and the normal to point of incidence all lie in the same plane.
2. The angle of incidence is equal to the angle of reflection.

Procedure :

1. Fix the white drawing paper on the drawing board using board pins.
2. Place the plane mirror vertically on the paper and use it to draw line M_1M_2.
3. On the midpoint O of M_1M_2 draw ON perpendicular to M_1M_2.
4. Draw a ray IO which makes an angle of $\angle 30°$ with the normal ON. This is the incident ray.
5. Place a plane mirror vertically along M_1M_2 and fix two all pins P_1 and P_2 on IO.
6. From the other side of the normal, see the images of pins P_1 and P_2 and fix two other pins P_3 and P_4 vertically so that pins P_3, P_4 and images of P_1 and P_2 appear collinear at their lower ends.
7. Now remove the pins and the mirror. Join the dots made by P_3 and P_4 and obtain the reflected ray RO.
8. Measure the angle of reflection $\angle$RON using a protactor.
9. Find the corresponding angle of reflection for different angles of incidence such as 30°, 35°, 40°, 45°, etc.



Reflection from a plane mirror



angle of incidence = angle of reflection

Observations

S.No.	Angle of incidence ∠ION = i	Angle of reflection ∠RON = r
1.	30°	30°
2.		
3.		
4.		
5.		

Results : 1. The incident ray, the reflected ray, and the normal to point of incidence all lie in the same plane.

2. The angle of incidence is equal to the angle of reflection.

Precautions : 1. Plane mirror should be vertical.

2. Pins should be kept vertical.

3. Distance between the two pins should be 3-4 cm..

4. Plane mirror should be clean.

5. The lower ends of the pins should be used for checking.

6. Mirror should not be shifted from the fixed line.

Experiment (Light: Reflection and refraction from a plane surface)

Objective : Finding the refractive index of a rectangular glass slab or establishing the laws of refraction

Essential Apparatus: Glass-slab, drawing-board, white drawing paper, pencil, board pins, Protractor and scale.

Principle : When light ray passes from a less dense to a more dense substance, it bends towards the normal and when it passes from denser to less dense medium it bends away from the normal. This is known as refraction and it follows the following laws:

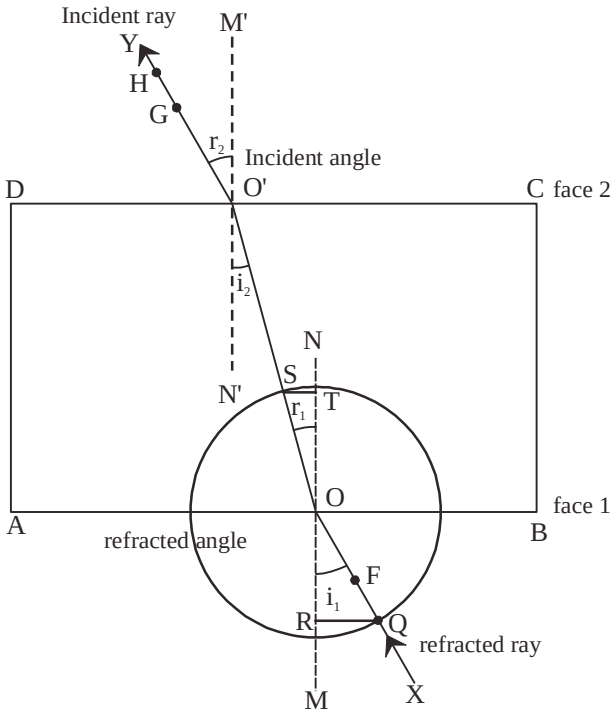
1. The incident ray, the refracted ray, and the normal to point of incidence all lie in the same plane.

2. The ratio of sine of angle of incidence to the sine of the angle of refraction is constant and it is known as the refractive index of the second medium with respect to the first.

$$_a\mu_g = \frac{\sin i}{\sin r}$$

The refractive index of glass with respect to air = $_a\mu_g = \frac{\text{length of perpendicular in air (QR)}}{\text{length of perpendicular in air (ST)}}$

Procedure : 1. Fix the white drawing paper on the drawing board using board pins.



2. Place the rectangular slab on the paper and draw its outline ABCD.
3. On any point O of AB, draw NM⊥AB. From O, draw a ray XO forming angle i with NM.
4. Place the slab properly and fix two pins E and F on XO. From the other side DC of the slab, try to see the image of the pins.
5. Fix two pins G and H upright so that images of E and F and pins G and H become collinear at their lower ends
6. Now remove the slab and the pins and draw emergent ray YO' passing through marks of G and H, which intersects DC at O'. Join OO'.
7. Keeping O at the centre, draw a circle of any radii which intersects OX at Q and OO' at S.
8. Draw QR⊥N_1O and ST⊥ON_2. Measure QR and ST.
9. Repeat at least 5 times for different values of i.
10. Calculate the refractive index for each reading and then find average refractive index value.

Observations

S.No.	**length of perpendicular in air, QR**	**length of perpendicular in glass, ST**	**refractive index** $\mu = \frac{QR}{ST}$
1.	..... cm.	..... cm.	
2.	..... cm.	..... cm.	
3.-	..... cm.	..... cm.	
4.	..... cm.	..... cm.	
5.	..... cm.	..... cm.	

Results : The refractive index of glass with respect to air is

Precautions : 1. Correct measurements should be indicated on the paper on the basis of the glass-slab.
2. Pins should be fixed vertically.
3. The images of first two pins and the remaining two pins should be in the same line.

4. Perpendiculars should be drawn carefully.
5. Angles should be properly measured.

Experiment (Light: Reflection and refraction from a curved surface)

Objective : Finding the focal length of a concave mirror using one-pin method.

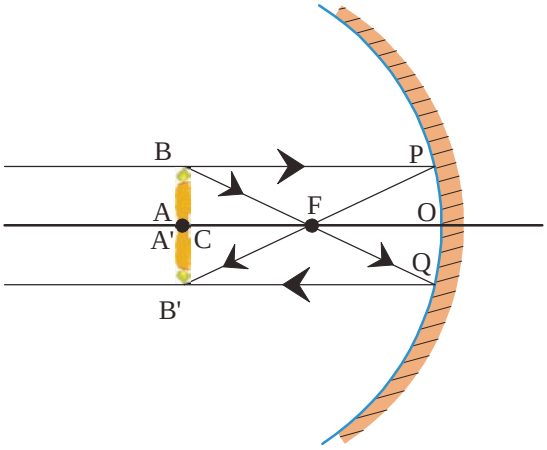
Essential Apparatus: light bench, concave mirror, pins etc.

Principle : When an object is placed at the centre of curvature of a concave mirror, a real image of same size is formed at the same place as the object. When distance between object and mirror is equal to radius of curvature, then focal length-

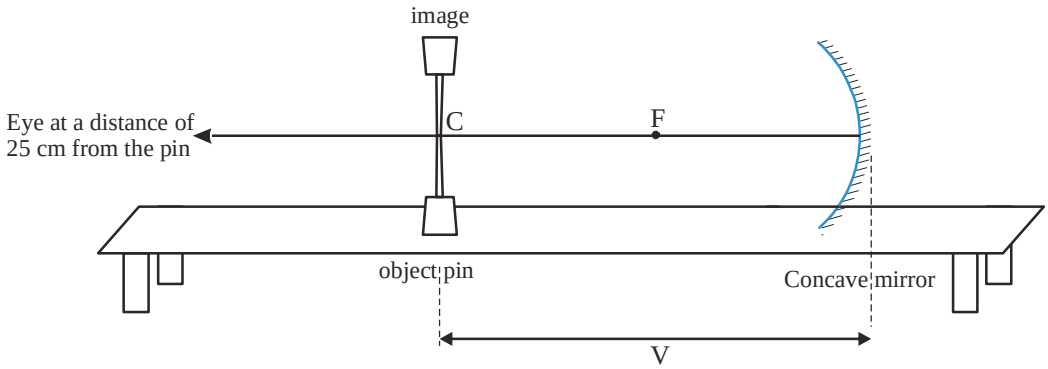
$$f = \frac{R}{2} \left(\text{focal length} = \frac{\text{radius of curvature}}{2} \right)$$

Procedure :

1. Fix the concave mirror vertically on one stand of the light bench and fix the bench horizontally flat.



2. On another clamp of the bench, fix a pin such that the tip of the pin is at the center of the mirror.
3. Move the pin along the bench in front of the mirror until a real inverted image of the pin is seen somewhere in front of the mirror. Adjust till there is no parallax between the pin and image.
4. Find the positions of the mirror and pin, the difference between the two will give the radius of curvature.
5. Repeat at least 4-5 times and take readings.



Finding focal length using one pin method

Observation table

S.No.	Position of mirror, M	Position of pin, P	Radius of curvature of mirror R = M - P
1.	..... cm	..... cm	..... cm
2.	..... cm	..... cm	..... cm
3.	..... cm	..... cm	..... cm
4.	..... cm	..... cm	..... cm

Average radius of curvature R = ……cm

Calculations : focal length $= \frac{\text{radius of curvature}}{2}$, focal length $f = \frac{R}{2} =$ .....cm

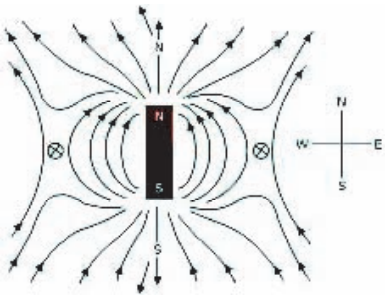
Results : The focal length of the concave mirror is = ……..cm

Precautions :
1. Pin and mirror should be aligned.
2. Parallax should be eliminated between object pin and its image.
3. The optical bench should be aligned flat horizontally before experiment.
4. Mirror should be clean and the room should have proper light.

Experiment (Magnetic effects of current)

Objective : Drawing magnetic force lines by aligning the North Pole of a bar magnet with geographic north (N→N).

Essential Apparatus: bar magnet, drawing-board, white drawing paper, board pins, compass etc.



Principle : The magnetic field lines due to a bar magnet are closed loops in which the tangent drawn at any point gives the direction of field at that point. The points where the magnetic force is zero are known as neutral points.

Procedure :
1. Fix the white drawing paper flat on the drawing board using board pins.
2. Place the compass on the paper and mark north-south direction on it. Draw a line joining the two points. Align the north pole of the bar magnet with the geographical north and trace the outline of the magnet on paper.
3. Place the compass at North pole of the magnet and make a dot next to the needle showing the direction the compass arrow points.
4. Move the compass so that the base of the arrow is at the dot you've just made. Now make a new mark where the tip of the arrow is pointing this time. Keep doing this until you reach the other end of the magnet. A curved line is obtained by joining the dots; this is the magnetic field line.

Similarly, draw several more magnetic lines of force each having arrows showing the direction from north to south. At the center of the lines, you will obtain a curved quadrilateral perpendicular to the axis. The neutral point lies in this quadrilateral.

5. Find the distance from the middle of the magnet to the neutral point.

Observations :

1. The distance from the mid-point of the magnet to the first neutral point is ….cm (this is average).
2. The distance from the mid-point of the magnet to the second neutral point is ….cm (this is average).

Results :

1. The distances of the neutral points from the mid-point of the bar magnet are nearly same. The points lie on the equatorial line of the bar magnet.
2. The field lines form a closed loop and the lines do not intersect each other at any point.

Precautions :

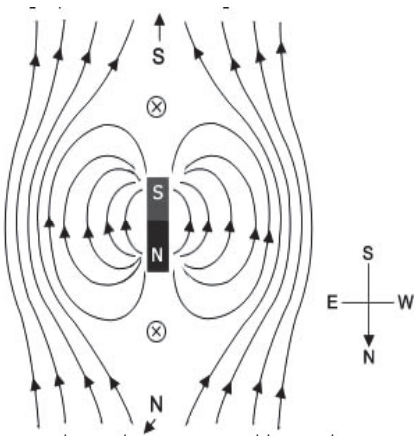
1. The drawing board and magnet should not be shifted during the experiment.
2. There should not be any other magnetic material around the experiment area.
3. Compass needle should move freely.
4. Magnet should not be very powerful.
5. More lines should be drawn near the neutral point so that it can be accurately located.

Experiment (Magnetic effects of current)

Objective : Drawing magnetic force lines by aligning the North Pole of a bar magnet with geographic south and finding neutral point (N→S).

Essential Apparatus: bar magnet, drawing-board, white drawing paper, board pins, compass etc.

Principle : Magnetic lines of force form a circular field in which the tangent drawn at any point gives the direction of field at that point. The points where the magnetic force is zero are known as neutral points. When the North pole of a bar magnet points towards the geographical south pole of the earth, the two neutral points lie on the axial line of the bar magnet such that they are equidistant from the centre of the bar magnet.



Procedure :

1. Fix the white drawing paper flat on the drawing board using board pins.
2. Place the compass on the paper and mark north-south direction on it. Draw a line joining the two points. Align the north pole of the bar magnet with the geographical south and trace the outline of the magnet on paper.
3. Place the compass at North pole of the magnet and make a dot next to the needle showing the direction the compass arrow points.
4. Move the compass so that the base of the arrow is at the dot you've just made. Now make a new mark where the tip of the arrow is pointing this time. Keep doing this until you reach the other end of the magnet. A curved line is obtained by joining the dots; this is the magnetic field line.

   Similarly, draw several more magnetic lines of force each having arrows showing the direction from north to south.
5. At the center of the lines, you will obtain a curved quadrilateral perpendicular to the axis. The neutral points lie in this quadrilateral in the direction of the axis of the magnet.
6. Find the distance from the middle of the magnet to the neutral points.

Observations :

1. The distance from the mid-point of the magnet to the first neutral point is ….cm (this is average).
2. The distance from the mid-point of the magnet to the second neutral point is ….cm (this is average).

Results :

1. The distances of the neutral points from the mid-point of the bar magnet are same. The points lie along the magnetic axis.
2. The field lines form a closed loop and the lines do not intersect each other at any point.

Precautions :

1. The drawing board and magnet should not be shifted during the experiment.
2. There should not be any other magnetic material around the experiment area.
3. Compass needle should move freely.
4. Magnet should not be very powerful.
5. More lines should be drawn near the neutral point so that it can be accurately located.